# LDA with Correlation over Words-Topic Distribution

## 1 Notations

- $K$ - number of topics;
- $M$ - number of documents in corpus;
- $N_d$ - number of words in document $d$;
- $V$ - number of words in vocabulary;
- $cnt_{dv}$ - the number of times word $w_{dv}$ appears in document $d$.

## 2 Generative process

- Choose distribution of topics in document $d$: $\theta_d \sim Dir(\alpha)$
- Choose distribution of words in topic $k$: $\beta'_k \sim N_V(\mu, \Sigma)$, with mapping to the simplex by $\beta_{kv} = \frac{exp\{\beta'_{kv}\}}{\sum_{v'=1}^{V} exp\{\beta'_{kv'}\}}$
- For $n \in \{1, ..., N\}$ in document $d$:
  - Choose topic $z_{dn} \sim Categorical(\theta_d)$
  - Choose word $w_{dn} \sim Categorical(\beta_{z_{dn}})$

## 3 Probability of the model

$$p(w, z, \theta, \beta; \alpha, \mu, \Sigma) = \prod_{k=1}^{K} p(\beta_k|\mu, \Sigma) \prod_{d=1}^{D} p(\theta_d|\alpha) \prod_{n=1}^{N} p(z_{dn}|\theta_d) p(w_{dn}|\beta_{z_{dn}})$$

## 4 Variational distribution

$$q(\theta, z, \beta) = \prod_{k=1}^{K} q(\beta_k|\lambda_k, \nu_k^2) \prod_{d=1}^{D} q(\theta_d|\gamma_d) \prod_{n=1}^{N} q(z_{dn}|\phi_{dn})$$

# 5 ELBO

$$L(\lambda, \nu^2, \gamma_d, \phi_d; \mu, \Sigma, \alpha) \geq \mathbb{E}_q[\log p(\theta_d|\alpha)] \quad (1)$$
$$+\mathbb{E}_q[\log p(z_d|\theta_d)] \quad (2)$$
$$+\mathbb{E}_q[\log p(\beta|\mu, \Sigma)] \quad (3)$$
$$+\mathbb{E}_q[\log p(w_d|z_d, \beta)] \quad (4)$$
$$-\mathbb{E}_q[\log q(\theta_d)] \quad (5)$$
$$-\mathbb{E}_q[\log q(z_d)] \quad (6)$$
$$-\mathbb{E}_q[\log q(\beta)] \quad (7)$$

1. $\mathbb{E}_q[\log p(\theta_d|\alpha)] = \log\Gamma(\sum_{k=1}^K \alpha_k) - \sum_{k=1}^K \log\Gamma(\alpha_k) + \sum_{k=1}^K (\alpha_k - 1)(\Psi(\gamma_{dk}) - \Psi(\sum_{k'=1}^K \gamma_{dk'}))$
   ["Latent Dirichlet Allocation", Eq. (15)]

2. $\mathbb{E}_q[\log p(z_d|\theta_d)] = \sum_{n=1}^N \sum_{k=1}^K \phi_{dnk}(\Psi(\gamma_{dk}) - \Psi(\sum_{k'=1}^K \gamma_{dk'}))$
   ["Latent Dirichlet Allocation", Eq. (15)]

3. $\mathbb{E}_q[\log p(\beta|\mu, \Sigma)] = \frac{1}{2}\log|\Sigma^{-1}| - \frac{V}{2}\log 2\pi - \frac{1}{2}\mathbb{E}_q[(\beta - \mu)^T\Sigma^{-1}(\beta - \mu)]$,
   where $\mathbb{E}_q[(\beta - \mu)^T\Sigma^{-1}(\beta - \mu)] = Tr(diag(\nu^2)\Sigma^{-1}) + (\lambda - \mu)^T\Sigma^{-1}(\lambda - \mu)$
   ["Correlated Topic Models", Eq. (8), (9)]
   $\mathbb{E}_q[\log p(\beta_k|\mu, \Sigma)] = \frac{1}{2}\log(\prod_{v=1}^V (1/\sigma_{kv}^2)) - \frac{V}{2}\log 2\pi - \frac{1}{2}\sum_{v=1}^V \nu_{kv}^2(1/\sigma_{kv}^2) - \frac{1}{2}\sum_{v=1}^V (\lambda_{kv} - \mu_{kv})(1/\sigma_{kv}^2)(\lambda_{kv} - \mu_{kv})$

4. $\mathbb{E}_q[\log p(w_d|z_d, \beta)] = \mathbb{E}_q[\log \beta_{z_{dn}, w_{dn}}] = \mathbb{E}_q[\log \prod_{v=1}^V \prod_{k=1}^K \beta_{kv}^{\mathbb{1}[v=w_{dn}, z_{dn}=k]}] = \sum_{v=1}^V \sum_{k=1}^K \mathbb{E}_q[\mathbb{1}[v = w_{dn}, z_{dn} = k]\log \beta_{kv}] = \sum_{v=1}^V \sum_{k=1}^K cnt_{dv}\phi_{dnk}\mathbb{E}_q[\log \beta_{kv}]$,
   where $\mathbb{E}_q[\log \beta_{kv}] = \mathbb{E}_q[\log \frac{exp\{\beta'_{kv}\}}{\sum_{v'=1}^V exp\{\beta'_{kv'}\}}] = \mathbb{E}_q[\beta'_{kv}] - \mathbb{E}_q[\log(\sum_{v'=1}^V exp\{\beta'_{kv'}\})] \geq \lambda_{kv} - \log\zeta_k - \zeta_k^{-1}(\sum_{v'=1}^V \mathbb{E}_q[exp\{\beta'_{kv'}\}]) + 1 = \lambda_{kv} - \log\zeta_k - \zeta_k^{-1}(\sum_{v'=1}^V exp\{\lambda_{kv'} + \frac{1}{2}\nu_{kv'}^2\}) + 1$

   To preserve the lower bound on the log probability, we upper bound the log normalizer with a Taylor expansion $f(a) + \frac{f'(a)}{1!}(x - a)$, where $f(.) = log(.)$ and $a = \zeta$,
   $\mathbb{E}_q[\log(\sum_{v'=1}^V exp\{\beta'_{kv'}\})] \leq \log\zeta_k + \zeta_k^{-1}(\sum_{v'=1}^V \mathbb{E}_q[exp\{\beta'_{kv'}\}]) - 1$
   The expected value of a log-normal random variable is $\mathbb{E}_q[exp\{\beta'_{kv'}\}] = exp\{\lambda_{kv'} + \frac{1}{2}\nu_{kv'}^2\}$.
   [Close to "Correlated Topic Models", Eq. (7), (11) and article "Log-normal distribution"]

5. $-\mathbb{E}_q[\log q(\theta_d)] = -\log\Gamma(\sum_{k=1}^K \gamma_{dk}) + \sum_{k=1}^K \log\Gamma(\gamma_{dk}) - \sum_{k=1}^K (\gamma_{dk}-1)(\Psi(\gamma_{dk}) - \Psi(\sum_{k'=1}^K \gamma_{dk'}))$
   ["Latent Dirichlet Allocation", Eq. (15)]

6. $-\mathbb{E}_q[\log q(z_d)] = -\sum_{n=1}^N \sum_{k=1}^K \phi_{dnk} \log \phi_{dnk}$
   ["Latent Dirichlet Allocation", Eq. (15)]

7. $-\mathbb{E}_q[\log q(\beta)] = \sum_{v=1}^V \sum_{k=1}^K \frac{1}{2}(\log \nu_{kv}^2 + \log 2\pi + 1)$
   ["Correlated Topic Models", Eq. (12)]
   $-\mathbb{E}_q[\log q(\beta_k)] = \sum_{v=1}^V \frac{1}{2}\log \nu_{kv}^2 + \frac{V}{2}\log 2\pi + \frac{V}{2}$

**E-step** Local (single document) optimization, these are distribution over topics for a document and topic assignments for words in a document. For each document, find the optimizing values of the variational parameters $\{\gamma_d, \phi_d\}$.
**M-step** Global (corpus wide) optimization, these are distribution over words in a topic and free variational parameter introduced through the Taylor expansion. Find the optimizing values of the parameters $\{\lambda_k, \nu_k^2, \zeta_k\}$.

For first iteration the model hyperparameters $\{\mu, \Sigma, \alpha\}$ can be seen as fixed and can be extracted into constants.

# 6 M-step

1. $\frac{dL}{d\lambda_{kv}} = (\cancel{\frac{1}{2}\log|\Sigma^{-1}|} - \cancel{\frac{V}{2}\log 2\pi} - \cancel{\frac{1}{2}Tr(diag(\nu^2)\Sigma^{-1})} - \frac{1}{2}(\lambda-\mu)^T\Sigma^{-1}(\lambda-\mu) + \sum_{v=1}^V \sum_{k=1}^K cnt_{dv}\phi_{dnk}(\lambda_{kv} - \cancel{\log\zeta_k} - \zeta_k^{-1}(\sum_{v'=1}^V exp\{\lambda_{kv'} + \frac{1}{2}\nu_{kv'}^2\}) + \cancel{1}) + \cancel{\sum_{v=1}^V \sum_{k=1}^K \frac{1}{2}(\log\nu_{kv}^2 + \log 2\pi + 1)})d\lambda_{kv} = (-\frac{1}{2}(\lambda_{kv}-\mu)(1/\sigma_{kv}^2)(\lambda_{kv}-\mu) + cnt_{dv}\phi_{dnk}(\lambda_{kv} - \zeta_k^{-1}exp\{\lambda_{kv} + \frac{1}{2}\nu_{kv}^2\}))d\lambda_{kv} = \underbrace{-(1/\sigma_{kv}^2)(\lambda_{kv}-\mu)}_{proof} + cnt_{dv}\phi_{dnk}(1 - \zeta_k^{-1}exp\{\lambda_{kv} + \frac{1}{2}\nu_{kv}^2\})$
   [Close to "Correlated Topic Models", Eq. (16)]

2. $\frac{dL}{d\nu_{kv}^2} = (\cancel{\frac{1}{2}\log|\Sigma^{-1}|} - \cancel{\frac{V}{2}\log 2\pi} - \frac{1}{2}Tr(diag(\nu^2)\Sigma^{-1}) - \cancel{\frac{1}{2}(\lambda-\mu)^T\Sigma^{-1}(\lambda-\mu)} + \sum_{v=1}^V \sum_{k=1}^K cnt_{dv}\phi_{dnk}(\cancel{\lambda_{kv}} - \cancel{\log\zeta_k} - \zeta_k^{-1}(\sum_{v'=1}^V exp\{\lambda_{kv'} + \frac{1}{2}\nu_{kv'}^2\}) + \cancel{1}) + \sum_{v=1}^V \sum_{k=1}^K \frac{1}{2}(\log\nu_{kv}^2 + \log 2\pi + 1))d\nu_{kv}^2 = (-\frac{1}{2}\nu_{kv}^2(1/\sigma_{kv}^2) - \frac{cnt_{dv}\phi_{dnk}}{\zeta_k}exp\{\lambda_{kv} + \frac{1}{2}\nu_{kv}^2\} + \frac{1}{2}\log\nu_{kv}^2)d\nu_{kv}^2 = -\frac{1}{2}(1/\sigma_{kv}^2) - \frac{cnt_{dv}\phi_{dnk}}{2\zeta_k}exp\{\lambda_{kv} + \frac{1}{2}\nu_{kv}^2\} + \frac{1}{2\nu_{kv}^2}$
   [Close to "Correlated Topic Models", Eq. (17)]

3. The derivative with respect to $\zeta_k$ has a maximum at $\hat{\zeta_k} = \sum_{v=1}^{V} exp\{\lambda_{kv} + \frac{1}{2}\nu_{kv}^2\}$.

   ["Correlated Topic Models", Eq. (14)]

Equations (1) and (2) is not amenable to analytic maximization, so we use a gradient algorithm to solve them.